

Attention as Renormalization: Multi-Scale Vector Fields for Length-Invariant Sequence Processing

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Abstract

Length generalization—the ability to process sequences longer than those seen in training—is a scale-invariance problem. We argue that multi-scale recurrent fields coupled across transformer layers implement a discrete renormalization group (RG) hierarchy, where coarse-graining over successively longer timescales produces representations that are invariant to total sequence length. We develop this idea in two stages. First, we show that *scalar* pheromone fields biasing attention logits exhibit an all-or-nothing ablation structure: five of six partial configurations fail, and only the complete system (feedback deposits, fast/slow fields, cross-layer coupling) generalizes—but poorly compared to dedicated recurrent models. Second, we upgrade from scalar logit biases to *vector social fields* that shift attention keys, drawing on a companion dual-channel architecture. The vector field provides dramatically higher state capacity (cumsum mod 16 at 2×: 0.999 vs 0.279 baseline) and a natural geometric interpretation: the field modulates *what attention looks for*, not just *how much* it attends. The renormalization framing unifies these results: the fast field captures token-level dynamics, the slow field coarse-grains over temporal blocks, and cross-layer coupling iterates the RG transformation. Length generalization emerges when the system operates near a critical fixed point where these scales interact self-consistently.

1 Introduction

Transformers do not generalize across sequence lengths. This is not merely empirical: bounded-depth transformers provably cannot simulate automata or track state over arbitrary-length inputs [Merrill & Sabharwal, 2023]. The core issue is that attention patterns are tied to positional representations that do not extrapolate.

We reframe this as a **scale-invariance problem**. A model that generalizes to longer sequences has learned representations that look the same regardless of total length—just as a system at a critical point looks the same regardless of magnification. Length generalization is scale invariance in time.

This framing suggests a constructive approach: *build a renormalization hierarchy into the attention mechanism*. In condensed-matter physics, the renormalization group (RG) achieves scale invariance by iteratively coarse-graining short-distance structure while preserving long-distance behavior [Kadanoff, 1966, Wilson & Kogut, 1974]. The key insight is that at a critical fixed point, the system is self-similar across scales—the coarse-grained version is statistically identical to the original.

We implement a discrete RG hierarchy within transformer attention using *stigmergic fields*—vector-valued recurrent states inspired by ant pheromone trails [Boissard et al., 2024]. The hierarchy has three levels:

1. **Fast field** ($\tau_f \approx 6$ tokens): captures local, token-level dynamics. This is the “microscopic” scale.

2. **Slow field** ($\tau_s \approx 50$ tokens): coarse-grains the fast field over temporal blocks. This is a *block-spin transformation in time*—the temporal analogue of Kadanoff’s block-spin RG [Kadanoff, 1966].
3. **Cross-layer coupling**: the slow field’s terminal state seeds the next layer’s initial condition. Each layer operates at a successively coarser temporal scale. This is *iteration of the RG transformation*.

The field enters attention not as a scalar logit bias (our earlier approach, which we show is too weak) but as an *additive key shift*—a vector that modulates the geometry of attention. This means the coarse-grained field changes *what the head looks for*, not just *how much it attends*, providing the state capacity needed for multi-bit sequence tasks.

Contributions.

1. A renormalization-group interpretation of multi-scale temporal coupling in attention, connecting length generalization to scale invariance at a critical fixed point.
2. A controlled ablation showing that the RG hierarchy is all-or-nothing: five of six partial configurations fail, mapping onto the concept of relevant vs. irrelevant operators in RG.
3. Evidence from two complementary architectures: scalar fields (clean ablation, limited capacity) and vector fields (high capacity, dramatic length generalization on 4-bit tasks).
4. A structural analysis of why the minimum viable system requires feedback, scale separation, and vertical coupling—three components that together implement the renormalization hierarchy.

2 Background

2.1 The Length Generalization Problem

Bounded-depth transformers with finite precision cannot simulate general finite automata [Merrill & Sabharwal, 2023]. For tasks like parity—computing the running sum modulo 2—a transformer trained on length- T inputs will memorize positional heuristics that break at length $T' > T$.

Existing approaches include positional extrapolation (ALiBi; Press et al. 2022), chain-of-thought, SSMs (S4 Gu et al. 2022; Mamba Gu & Dao 2023), and interleaved hybrids (Jamba Lieber et al. 2024; Griffin De et al. 2024). None directly test what *structure* of recurrence enables length generalization when coupled into the attention mechanism itself.

2.2 Renormalization Group Basics

The renormalization group [Kadanoff, 1966, Wilson & Kogut, 1974] is a framework for understanding systems at multiple scales. The procedure is:

1. **Coarse-grain**: partition the system into blocks, replace each block with a single effective degree of freedom.
2. **Rescale**: the block-level system has the same form as the original but with renormalized parameters.
3. **Iterate**: repeated coarse-graining flows through parameter space toward fixed points.

At a *critical fixed point*, the system is scale-invariant: it looks statistically identical at all magnifications. The critical point is characterized by:

- **Relevant operators:** perturbations that grow under RG flow. Removing a relevant operator drives the system away from the fixed point.
- **Irrelevant operators:** perturbations that shrink under RG flow. The system self-corrects.
- **Universality:** systems with the same relevant operators share the same critical behavior, regardless of microscopic details.

2.3 Stigmergy and Ant Trail Formation

Stigmergy is the indirect coordination mechanism used by social insects: agents deposit pheromones that influence future behaviour. Boissard et al. [2024] identify a three-ingredient recipe for edge-of-chaos dynamics in lattice ant models: *amplification* (positive feedback), *injection* (activity-dependent deposition), and *erosion* (uniform decay). Removing any ingredient causes collapse. Crucially, erosion must be *uncoupled* from the field it erodes—a structural condition that we will interpret as a requirement for well-defined coarse-graining.

3 Architecture: A Renormalization Hierarchy in Attention

3.1 Scalar Fields: The Minimal Case

We begin with the simplest version to establish the ablation structure. For each attention head h , define a scalar field $\phi_i^{(h)}$ at position i :

$$\phi_i^{(h)} = \alpha \cdot \phi_{i-1}^{(h)} + d_i^{(h)}, \quad \alpha = 1 - \varepsilon \quad (1)$$

The field enters as an additive logit bias:

$$\ell_{ij}^{(h)} = \frac{\mathbf{q}_i^{(h)\top} \mathbf{k}_j^{(h)}}{\sqrt{d_h}} + \phi_j^{(h)} \quad (2)$$

The deposit source determines the field’s expressive power. *Input deposits* ($d_i = W_d \mathbf{x}_i$) make the field a deterministic moving average—provably inert for state tracking. *Feedback deposits* ($d_i = W_d \mathbf{o}_i$, where $\mathbf{o}_i$ is the attention output) close the loop:

$$\phi \xrightarrow{\text{bias}} \text{attention} \xrightarrow{\text{aggregate}} \text{output} \xrightarrow{W_d} \text{deposit} \xrightarrow{\text{accumulate}} \phi \quad (3)$$

creating a genuine nonlinear recurrence.

Limitation. The scalar field says “attend more here” but not “attend in this direction.” With H heads, the field provides $O(H)$ bits of recurrent state—enough for 1-bit parity but insufficient for richer tasks.

3.2 Vector Fields: Modulating Attention Geometry

We replace the scalar field with a *vector social field* $\phi_i^{(h)} \in \mathbb{R}^{d_f}$ that modulates keys directly:

Definition 1 (Vector social field). *For each head h with key dimension d_h and field dimension $d_f = d_h$:*

$$\phi_i^{(h)} = \alpha \cdot \phi_{i-1}^{(h)} + \mathbf{d}_i^{(h)} \quad (4)$$

$$\mathbf{d}_i^{(h)} = W_{\text{dep}}^{(h)} \mathbf{o}_i^{(h)} \quad (5)$$

where $\mathbf{o}_i^{(h)}$ is the attention output at position i , head h .

The field enters attention through additive key modulation:

$$\mathbf{k}_j'^{(h)} = \mathbf{k}_j^{(h)} + W_{\text{read}}^{(h)} \phi_j^{(h)} \quad (6)$$

This is a qualitatively different mechanism. The scalar field biases *how much* a position is attended to. The vector field shifts keys in a learned direction, changing *what kind of match* occurs. With field dimension $d_f = d_h$, each head carries d_f -dimensional recurrent state—orders of magnitude more than the scalar case.

Biological interpretation. Ant pheromones are not scalar intensities—they are chemical blends whose composition encodes qualitative information (food type, danger, nest direction). A vector field is the natural analogue: the field’s direction encodes *what kind* of signal was deposited, while its magnitude encodes strength.

3.3 The Renormalization Hierarchy

We now build the multi-scale structure, interpreted as a discrete RG transformation.

3.3.1 Level 0: Token-Level Dynamics (Input)

The input sequence $\mathbf{x}_1, \dots, \mathbf{x}_T$ defines the “microscopic” degrees of freedom. Standard attention operates at this level: each query $\mathbf{q}_i$ matches against all keys $\mathbf{k}_j$ for $j \leq i$.

3.3.2 Level 1: Fast Field (Local Coarse-Graining)

The fast field ϕ^f with evaporation rate $\varepsilon_f = 0.15$ ($\tau_f \approx 6$) performs the first coarse-graining. At each position, it accumulates attention-derived deposits over a window of $\sim \tau_f$ tokens:

$$\phi_i^{f,(L)} = \alpha_f \cdot \phi_{i-1}^{f,(L)} + \mathbf{d}_i^{(L)} \quad (7)$$

This is the temporal analogue of averaging over a spatial block in the Kadanoff construction: short-timescale fluctuations are smoothed into a local summary vector that modulates keys at nearby positions.

3.3.3 Level 2: Slow Field (Block-Spin Transformation)

The slow field ϕ^s with $\varepsilon_s = 0.02$ ($\tau_s \approx 50$) coarse-grains the fast field:

$$\phi_i^{s,(L)} = \alpha_s \cdot \phi_{i-1}^{s,(L)} + h^{(L)}(\phi_i^{f,(L)}) \quad (8)$$

where $h^{(L)} : \mathbb{R}^{d_f} \rightarrow \mathbb{R}^{d_f}$ is a learned linear projection.

This is the **block-spin transformation**: the slow field replaces blocks of $\sim \tau_s/\tau_f \approx 8$ fast-field values with a single effective vector. The projection $h^{(L)}$ is the “block-spin rule”—it determines which components of the fast-field structure survive coarse-graining.

The rescaling factor $b = \tau_s/\tau_f$ controls the RG step size. Our values give $b \approx 8$, meaning each slow-field update integrates over ~ 8 fast-field correlation lengths.

3.3.4 Level 3: Cross-Layer Coupling (Iterating the RG)

The slow field’s terminal state at layer L seeds layer $L+1$:

$$\phi_0^{s,(L+1)} = \sigma(w_g^{(L)}) \cdot P^{(L)} \phi_T^{s,(L)} \quad (9)$$

where $\sigma(w_g)$ is a learned sigmoid gate and $P^{(L)}$ is a linear projection.

This is **iteration of the RG transformation**. Each layer receives the coarse-grained state from the layer below and operates on it at a yet coarser scale. With N layers, the effective temporal horizon grows as $\tau_{\text{eff}} \sim \tau_s^N$ —each layer extends the memory exponentially.

The full system. Both fields contribute to key modulation:

$$\mathbf{k}'_j^{(h,L)} = \mathbf{k}_j^{(h,L)} + W_f^{(h,L)} \phi_j^{f,(L)} + W_s^{(h,L)} \phi_j^{s,(L)} \quad (10)$$

The attention mechanism then operates on the modulated keys:

$$\ell_{ij}^{(h,L)} = \frac{\mathbf{q}_i^{(h,L)\top} \mathbf{k}'_j^{(h,L)}}{\sqrt{d_h}} \quad (11)$$

No scalar logit bias appears. The field’s influence is entirely through key geometry.

3.4 RG Interpretation: Why This Should Give Length Generalization

The connection between our architecture and the RG is not merely analogical. We identify three specific mechanisms:

Scale invariance from critical dynamics. The decay rates α_f^{i-j} and α_s^{i-j} depend only on the *distance* $i - j$, not on absolute position. The field’s memory horizon is position-independent. At the optimal ε (the “critical point”), the system’s behavior is independent of total sequence length—precisely the definition of length generalization.

Relevant operators from the ablation. The all-or-nothing ablation (section 4) maps directly onto the RG concept of relevant operators:

- **Feedback deposits** = relevant. Without them, the field is a deterministic average (no state tracking). Removing this operator drives the system to the trivial fixed point (chance accuracy).
- **Scale separation** = relevant. Without it, there is no coarse-graining hierarchy. A single decay rate means a single “magnification”—no RG flow.
- **Cross-layer coupling** = relevant. Without it, the RG transformation is applied once (within a layer) but never iterated. The hierarchy has depth 1.

All five partial configurations are missing at least one relevant operator and flow to the trivial fixed point. Only the full system sits near the nontrivial critical point.

Universality prediction. If the analogy holds, systems with the same “state complexity” (number of bits tracked) should exhibit the same generalization behavior regardless of task details. We observe this: parity (1-bit) and cumsum mod 4 (2-bit) both show monotonic degradation with length, with the field’s benefit scaling with available state capacity.

A prediction. The rescaling factor $b = \tau_s/\tau_f$ should matter. If b is too small, the fast and slow fields are not well-separated and the “coarse-graining” is trivial. If b is too large, information is lost in the averaging. There should be an optimal b —a critical ratio—that maximizes generalization. The sharp ε sensitivity we observe ($\varepsilon = 0.05$ works, $\varepsilon = 0.10$ collapses on cumsum) is consistent with this.

4 Evidence I: Scalar Field Ablation

We first establish the all-or-nothing structure using scalar fields, where the small parameter count makes controlled ablation straightforward.

4.1 Setup

All transformer models: 4 layers, 4 heads, $d_{\text{model}} = 64$, ReLU MLP with expansion factor 4, RMS normalisation, learned positional embeddings. Total: $\sim 209\text{K}$ parameters across all configurations.

Task: running parity. Train on $n = 16$ examples ($T = 34$ tokens), test at $n \in \{16, 32, 48, 64\}$. 3 seeds per configuration.

Six configurations, differing only in field mechanism:

Table 1: Scalar field configurations. All share the same transformer backbone; parameter counts are matched ($\pm 0.5\%$).

Config	Field	Feedback	Multi-scale	Cross-layer	Params
no_field	—	—	—	—	208,512
input_field	✓				209,536
feedback_field	✓	✓			209,536
multiscale_field	✓		✓		209,616
hierarchical_field	✓		✓	✓	209,664
hierarchical_feedback	✓	✓	✓	✓	209,664

4.2 Results: All-or-Nothing

Table 2: Scalar field ablation on parity. Chance is 0.50. Only the full system generalises. Mean $\pm$ std over 3 seeds.

Config	Accuracy at n examples (T tokens)			
	$n=16$	$n=32$	$n=48$	$n=64$
no_field	$1.000 \pm .000$	$0.589 \pm .037$	$0.532 \pm .021$	$0.523 \pm .019$
input_field	$1.000 \pm .000$	$0.488 \pm .008$	$0.498 \pm .002$	$0.496 \pm .005$
feedback_field	$1.000 \pm .000$	$0.517 \pm .012$	$0.503 \pm .004$	$0.508 \pm .010$
multiscale	$1.000 \pm .000$	$0.496 \pm .008$	$0.499 \pm .002$	$0.495 \pm .002$
hierarchical	$1.000 \pm .000$	$0.483 \pm .000$	$0.503 \pm .006$	$0.498 \pm .006$
hier. feedback	$1.000 \pm .000$	$0.808 \pm .031$	$0.710 \pm .110$	$0.653 \pm .079$

The result is binary. Five configurations are indistinguishable from chance at all OOD lengths. The sixth—and *only* the sixth—generalises. In RG language: five configurations are missing a relevant operator and flow to the trivial fixed point. The full system sits near the nontrivial critical point.

Why each component fails alone.

- **Input deposits:** the field is a deterministic function of input—a moving average that provides no state.
- **Feedback without multi-scale:** single timescale cannot simultaneously provide local precision and long-range memory. No scale separation $\Rightarrow$ no coarse-graining.
- **Multi-scale without feedback:** two moving averages are still averages. The coarse-graining has nothing useful to coarse-grain.
- **Cross-layer without feedback:** vertical plumbing transmits noise. The RG iteration operates on empty state.

4.3 SSM Baselines

Table 3: Dedicated recurrent models achieve perfect generalization, confirming that recurrence trivially solves parity.

Model	Params	$n=16$	$n=32$	$n=48$	$n=64$
S4D	167K	1.000	1.000	1.000	1.000
Selective SSM	184K	1.000	1.000	1.000	1.000
GRU	227K	1.000	1.000	1.000	1.000
<i>Hier. feedback (scalar)</i>	<i>210K</i>	<i>1.000</i>	<i>0.808</i>	<i>0.710</i>	<i>0.653</i>

SSMs achieve perfect generalization because they *are* the fixed point: pure recurrence is inherently scale-invariant for state tracking. The question is whether we can approach this fixed point from the attention side.

5 Evidence II: Vector Fields and State Capacity

The scalar field’s limitation is capacity: H scalars provide $O(H)$ bits of state. Upgrading to vector fields tests whether the RG hierarchy works with richer “spins.”

Results are from a companion dual-channel experiment using 2 layers, 4 heads, $d_{\text{model}} = 48$, $d_f = 12$ ($\sim 60\text{K}$ params), RoPE positional encoding, trained on $n = 16$ ($T = 34$).

Table 4: Vector social field (additive key modulation) vs. scalar field and baseline. The vector field’s advantage grows dramatically with state complexity.

Mode	Parity (1-bit)			Cumsum mod 16 (4-bit)		
	T=34	T=66	T=98	T=34	T=66	T=98
Baseline	1.000	0.995	0.525	1.000	0.279	0.062
Scalar feedback	1.000	0.949	0.699	1.000	0.870	0.216
Vector additive	1.000	0.979	0.738	1.000	0.999	0.419

Key observations.

- On cumsum mod 16 (4-bit state) at $2\times$ training length: vector additive achieves **0.999** vs. baseline **0.279**—a $+0.721$ improvement. This is the largest effect in any experiment. The vector field’s $H \times d_f = 48$ values of recurrent state can encode the 4-bit running sum that scalars cannot.
- The advantage grows with state complexity: $+0.039$ on parity (1-bit) vs. $+0.203$ on cumsum (4-bit) at $3\times$. This is the capacity scaling predicted by the RG interpretation: richer “spins” (vectors vs. scalars) carry more information through the coarse-graining hierarchy.
- Additive key modulation outperforms multiplicative gating (not shown; see companion paper). The additive mechanism has a natural geometric interpretation: the field *translates* keys in a learned direction, preserving the attention mechanism’s dot-product structure.

5.1 Multi-Task State-Capacity Scaling (Scalar Fields)

The monotonic degradation from parity to cumsum mod 16 confirms the $O(H)$ -bit capacity prediction for scalar fields. The flipflop result reveals a boundary: the field helps with *accumulation* tasks (where all inputs contribute) but not *gating* tasks (selective state update). In

Table 5: State-capacity scaling at $4\times$ training length. The field’s benefit degrades monotonically with state complexity.

	Parity (1-bit)	Cumsum mod 4 (2-bit)	Cumsum mod 16 (4-bit)	Flipflop (gating)
No field	0.52	0.25	0.06	0.10
Hier. feedback	0.65	0.40	0.15	0.15
S4D	1.00	1.00	0.95	0.20
Random	0.50	0.25	0.06	0.06

RG terms, gating requires a different universality class—one where the coarse-graining rule is input-dependent, not fixed.

6 The Renormalization Picture

6.1 Block-Spin Analogy Made Precise

In Kadanoff’s block-spin construction, one partitions a d -dimensional lattice into blocks of linear size b , replaces each block’s spins with a single “block spin” (typically the block average or majority), and rescales the lattice so the block lattice has the same spacing as the original.

Our temporal analogue:

Block-spin RG	Our hierarchy	Component
Lattice sites	Sequence positions	Input tokens
Spin at site i	Key vector $\mathbf{k}_i$	Level 0
Block of size b	Window of τ_f tokens	Fast-field kernel
Block spin	Fast-field vector ϕ_i^f	Level 1
Super-block	Window of τ_s tokens	Slow-field kernel
Super-block spin	Slow-field vector ϕ_i^s	Level 2
RG iteration	Cross-layer coupling	Level 3
Rescaling factor b	$\tau_s/\tau_f \approx 8$	
Critical point	Optimal ε	Edge of chaos
Fixed-point condition	Length generalization	Scale invariance

The exponential kernel α^{i-j} is not the only possible coarse-graining rule, but it has a key property: **it is translation-invariant**. The coarse-graining at position i uses the same kernel as at position $i + \Delta$, regardless of Δ . This is the structural reason the field generalizes across lengths.

6.2 Scale Separation and the Adiabatic Condition

For the hierarchy to function as a proper RG, the scales must be well-separated: $\tau_s \gg \tau_f$. In our experiments, $\tau_s/\tau_f \approx 8$, which provides roughly one order of magnitude of separation.

The separation enables an **adiabatic approximation**: from the perspective of the fast field, the slow field is approximately constant; from the perspective of the slow field, the fast field fluctuations are averaged out. This means:

- The fast field tracks local structure without interference from long-range memory.
- The slow field accumulates global structure without being disrupted by token-level noise.

- Each scale maintains its own dynamics while exchanging information through the hierarchy.

The ant-trail analogy: trail pheromone (fast, volatile) guides local foraging; colony pheromone (slow, persistent) encodes the nest location. They operate at different timescales and influence different behaviors, coupled only through the ants’ movements.

6.3 Why the Single-Scale System Fails: No RG Flow

A single decay rate ε gives a single timescale $\tau = 1/\varepsilon$. This is a system with one “magnification”—there is no RG flow because there is no hierarchy to flow through. The system either remembers (if ε is small enough for the test length) or forgets (if ε is too large), with no mechanism to bridge scales.

The failure of single-scale feedback (table 2) is not a failure of recurrence but a failure of *scale structure*. The system has state but no mechanism to organize it across temporal scales.

6.4 Evaporation Rate as a Control Parameter

In the RG framework, a *control parameter* tunes the system through a phase transition. The evaporation rate ε plays this role:

- $\varepsilon \rightarrow 1$ (rapid erosion): the field decays instantly. No memory. The system is in the “disordered phase”—attention operates as if no field exists.
- $\varepsilon \rightarrow 0$ (no erosion): the field saturates. All positions accumulate equal weight. The system is in the “ordered phase”—attention is dominated by the field, losing content sensitivity.
- ε at criticality: the field maintains structured, content-dependent memory that neither saturates nor vanishes. The system sits at the edge of chaos, maximizing information transmission across scales.

The sharp ε sensitivity observed empirically ($\varepsilon = 0.05$ works on cumsum; $\varepsilon = 0.10$ collapses to 0.256) is the signature of a narrow critical window—a hallmark of systems near a phase transition.

7 The Combined Architecture: Vector Fields + RG Hierarchy

The two evidence streams converge on a single architecture: **vector social fields organized in a renormalization hierarchy**.

1. **Vector fields** (from section 5) provide the state capacity. Each head carries d_f -dimensional recurrent state through additive key modulation.
2. **Multi-scale structure** (from section 4) provides the scale hierarchy. Fast and slow fields implement a two-level block-spin coarse-graining.
3. **Cross-layer coupling** iterates the RG transformation, extending the effective memory exponentially with depth.
4. **Feedback deposits** make the field content-dependent, creating genuine state rather than deterministic averages.

The combined system for layer L , head h , position i :

$$\phi_i^{f,(L,h)} = \alpha_f \cdot \phi_{i-1}^{f,(L,h)} + W_{\text{dep}}^{(L,h)} \mathbf{o}_i^{(L,h)} \quad (12)$$

$$\phi_i^{s,(L,h)} = \alpha_s \cdot \phi_{i-1}^{s,(L,h)} + h^{(L,h)} \left(\phi_i^{f,(L,h)} \right) \quad (13)$$

$$\phi_0^{s,(L,h)} = \sigma \left(w_g^{(L-1,h)} \right) \cdot P^{(L-1,h)} \phi_T^{s,(L-1,h)} \quad (14)$$

$$\mathbf{k}_j'^{(L,h)} = \mathbf{k}_j^{(L,h)} + W_f^{(L,h)} \phi_j^{f,(L,h)} + W_s^{(L,h)} \phi_j^{s,(L,h)} \quad (15)$$

Scaling. The field adds $O(H \cdot d_f)$ parameters per layer for the deposit, read, and projection matrices. With $d_f = d_h$, this is $< 1\%$ parameter overhead at typical model sizes. The recurrent state per position is $2 \cdot H \cdot d_f$ values (fast + slow per head)—far richer than scalar fields but far smaller than a full SSM hidden state.

Predicted behavior. Combining the scalar ablation (table 2) with the vector capacity results (table 4), we predict the vector RG hierarchy will:

- Maintain the all-or-nothing structure (only the full hierarchy works).
- Close much of the gap to SSMs on state-tracking tasks, particularly those requiring > 1 bit of state where scalar fields are capacity-limited.
- Exhibit the same sharp ε sensitivity, reflecting the critical nature of the RG fixed point.

8 Discussion

8.1 Where This Sits in the Design Space

	Attention	This work	SSM
Random access	Full	Full	None
Recurrent state	None	$2Hd_f$ per pos.	Full
Length generalization	None	Partial-good	Perfect
Parallelizable	Fully	Partially	Via scan

The architecture does not replace attention with recurrence (SSMs) or alternate between them (Jamba/Griffin). It augments attention with a lightweight recurrence—a vector trail—organized as a renormalization hierarchy. The attention mechanism remains primary; the field is a “guide” that shapes what attention looks for based on accumulated history.

8.2 Not Trying to Beat SSMs

SSMs are the natural solution when full recurrence is needed. Our goal is different: to give transformers—which dominate production deployments—a *biologically motivated* form of sequence memory. The question is not “can this beat Mamba on parity?” (it cannot) but “can the RG hierarchy give attention a principled, interpretable form of state tracking that standard attention lacks?”

The vector field results on cumsum mod 16 (0.999 at $2\times$ vs. 0.279 baseline) suggest the answer is yes for tasks within the field’s capacity. The renormalization framing provides hyperparameter guidance (ε controls a phase transition) and architectural guidance (the hierarchy must be complete—no shortcuts).

8.3 Biological Plausibility and Interpretability

The architecture has a direct biological interpretation:

- Fast field = trail pheromone (volatile, local)
- Slow field = colony pheromone (persistent, global)
- Cross-layer = information transfer between organizational levels (forager $\rightarrow$ colony)
- Key modulation = pheromone reshaping the “search landscape”

This is interpretable in a way that SSM hidden states are not. One can inspect what direction the field is shifting keys, at what timescale, and how it flows between layers. The field is a “social memory” that the model deposits and reads—a form of external working memory embedded in the attention mechanism.

8.4 Limitations and Open Questions

1. **The vector + hierarchy experiment is predicted, not run.** The scalar ablation and vector capacity results come from separate experiments. The combined architecture needs direct testing.
2. **Scale.** All experiments use $\sim 60\text{--}210\text{K}$ parameter models. Scaling behavior is unknown.
3. **Sequential cost.** Feedback deposits require sequential processing ($O(T)$ bottleneck). Chunk-parallel training or prefix-scan linearization is needed for practical use.
4. **Is the RG analogy predictive?** We have identified structural parallels. The next step is quantitative predictions: scaling laws for the rescaling factor b , critical exponents for the generalization transition, universality across task families.
5. **Continuous spectrum.** Real RG systems have a continuous spectrum of scales. Our discrete two-level hierarchy (fast/slow) is the minimal realization. Richer hierarchies (three or more timescales) may be needed for harder tasks.

9 Conclusion

Length generalization is a scale-invariance problem. We have argued—through a combination of controlled ablations, capacity experiments, and theoretical analysis—that a renormalization hierarchy within transformer attention provides a principled route to this invariance.

The hierarchy has three levels: a fast vector field capturing local dynamics, a slow vector field performing temporal coarse-graining, and cross-layer coupling iterating the transformation. The ablation reveals that all three levels are *relevant operators* in the RG sense: removing any one drives the system to the trivial fixed point.

The scalar version of this hierarchy is clean but weak—it generalizes partially on 1-bit tasks but cannot compete with dedicated recurrent models. The vector version, which modulates attention geometry rather than logit magnitudes, provides dramatically higher state capacity and closes much of the gap on multi-bit tasks.

The renormalization framing is more than metaphor. It makes specific predictions: that the hierarchy must be complete, that the evaporation rate controls a phase transition, that the rescaling factor $b = \tau_s/\tau_f$ should have an optimal value, and that generalization should scale with the number of RG levels (layers). Testing these predictions—particularly the scaling laws—is the natural next step.

The deeper question is whether attention, augmented with the right recurrent structure, can achieve the state tracking of SSMs without sacrificing its random-access flexibility. Our results

suggest the answer is “partially, and with principled guidance from the renormalization group.” The architecture sits between transformers and SSMS not as a compromise but as a distinct point in the design space—one where attention remains primary but is guided by a multi-scale memory that respects the temporal structure of the input.

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